6TH GRADE MATH · UNIT 3: RATIONAL NUMBERS

Weather Station Watch

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run a winter weather station. They order and compare positive and negative temperatures, describe elevations above and below sea level, use absolute value for distances and changes, and map stations on the coordinate plane.

The context makes the meaning of 0, opposites, and absolute value concrete.

Standards Addressed

- 6.NS.C.5 Use positive and negative numbers in real-world contexts
- 6.NS.C.6 Rational numbers on the number line and coordinate plane
- 6.NS.C.7 Order and absolute value of rational numbers
- 6.NS.C.8 Distances between points with a shared coordinate

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a real weather map. Part 1 ordering temperatures on a number line.
Day 2	Part 2 elevation and absolute value.
Day 3	Part 3 coordinate map and the weather report.

Materials

- Student handout and Station Cards
- Grid paper
- Colored pencils
- Optional: local weather data from the past week

How the Three Versions Differ

Level	What changes
CP2	Integers only, ordering on a number line, absolute value as distance, and plotting points.
CP1	Fractions and decimals, changes across 0, and distances on the coordinate plane with a storm path.
Honors	Close decimals and fractions, means of signed data, reasoning about $\|a\|$ vs. a, and area and perimeter of a radar rectangle.

Watch For (Common Misconceptions)

- Thinking −12 > −3 because 12 > 3.
- Subtracting absolute values when points are on opposite sides of 0.
- Switching x and y when plotting.

Extension Ideas

- Collect a week of real temperatures and graph them.
- Add a seventh station and predict its weather.
- Compare Fahrenheit and Celsius readings below zero.

Answer Key – CP2

Part 1: Read the Weather

#	Task	Answer
1	Write the low temperature of each station. Which station was coldest?	Hilltop −12, Harbor 6, School −3, Lakeside −7, Farm 2, Valley −9. Hilltop (−12°F) was coldest.
2	Order the six low temperatures from least to greatest.	−12, −9, −7, −3, 2, 6
3	Which stations are below sea level? How do you know?	Harbor, Lakeside, and Valley; their elevations are negative.
4	What is the opposite of −7? Of 6?	7 and −6

Part 2: Distance from Zero

#	Task	Answer
5	Find $\|-12\|$, $\|6\|$, and $\|-40\|$.	12, 6, 40
6	Which station is farthest from sea level, up or down?	Hilltop, 420 ft above sea level ($\|420\| = 420$ is the greatest).
7	Valley is at −40 ft and Harbor is at −15 ft. Which is lower? How many feet apart are they?	Valley is lower; 40 − 15 = 25 ft apart.
8	The temperature went from −3°F to 5°F. How many degrees did it rise?	$\|-3\| + 5 = 8$ degrees

Part 3: Map the Stations

#	Task	Answer
9	Plot all six stations on a coordinate grid. Label each one.	Points: Hilltop (3, 4), Harbor (−2, −3), School (0, 0), Lakeside (−4, 2), Farm (5, −1), Valley (3, −4).
10	Which quadrant is each station in? (School is at the origin.)	Hilltop I, Lakeside II, Harbor III, Farm IV, Valley IV, School origin.
11	Hilltop (3, 4) and Valley (3, −4) are on the same line. How many miles apart are they?	4 + 4 = 8 miles

Reflection

#	Prompt	Look for
R1	Why is −12 less than −3 even though 12 is greater than 3?	−12 is farther left of 0 on the number line, so it is less.
R2	How does absolute value help describe distance?	Distance is never negative; absolute value tells how far a number is from 0 regardless of direction.

Answer Key – CP1

Part 1: Temperature Log

#	Task	Answer
1	Order the readings from least to greatest.	$-5.5, -3.5, -\frac{3}{4}, 0, 1\frac{1}{2}, 2.25$
2	Which reading is closest to 0? Which is farthest? Use absolute value.	Closest: 0 (then $-\frac{3}{4}$). Farthest: −5.5 ($\lvert -5.5 \rvert = 5.5$).
3	Write an inequality that compares −3.5 and $-\frac{3}{4}$, and explain it in words.	$-3.5 < -\frac{3}{4}$; −3.5°F is colder than $-\frac{3}{4}$°F.
4	How many degrees warmer was the warmest reading than the coldest?	2.25 − (−5.5) = 7.75 degrees

Part 2: Elevation & Change

#	Task	Answer
5	Order the stations by elevation from lowest to highest.	Valley −40, Harbor −15, Lakeside −6, Farm 35, School 80, Hilltop 420
6	How many feet higher is Hilltop than Valley?	420 + 40 = 460 ft
7	A weather balloon starts at Harbor and rises 115 ft. What is its elevation? Is it above or below School?	−15 + 115 = 100 ft; above School (80 ft).
8	Explain why "Lakeside is at −6 ft" and "Lakeside is 6 ft below sea level" mean the same thing.	The negative sign shows the direction (below 0), and 6 is the distance from sea level.

Part 3: Storm Tracker

#	Task	Answer
9	Plot the stations. Find the distance from Hilltop (3, 4) to Valley (3, −4). Why can you NOT find the distance from Lakeside (−4, 2) to Hilltop by counting along one grid line?	Hilltop and Valley share x = 3: $\lvert 4 \rvert + \lvert -4 \rvert = 8$ miles. Lakeside and Hilltop share neither coordinate, so they are not on one horizontal or vertical line.
10	The storm starts at (−5, 2), moves 9 miles east, then 6 miles south. Where does it end? Which station is it closest to?	(4, −4); closest to Valley (3, −4), 1 mile away.
11	Reflect Farm (5, −1) across the x-axis and across the y-axis. Name the new points and their quadrants.	Across x-axis: (5, 1), Quadrant I. Across y-axis: (−5, −1), Quadrant III.

Reflection

#	Prompt	Look for
R1	How is finding a change in temperature like finding a distance on a number line?	Both measure the gap between two numbers; when the numbers have opposite signs you add their absolute values.
R2	Why can you count units to find distance only when two points share a coordinate?	Then they lie on the same horizontal or vertical line, so the distance is the difference of the other coordinate.

Answer Key – Honors

Part 1: Data Detective

#	Task	Answer
1	Order the readings from coldest to warmest. Show how you compared them.	-4.35, $-4\frac{1}{3}$ ($-4.333\ldots$), -4.3, $-\frac{17}{4}$ (-4.25), $-4\frac{1}{6}$ (-4.1667)
2	Find the mean of 6, −12, −3, −7, 2, and −9 (the station lows). Interpret the sign.	Sum −23 ÷ 6 ≈ −3.83°F; the average low was below 0°F.
3	Is it possible for $\|a\| < \|b\|$ but $a > b$? Give an example from the readings or explain why not.	Yes. Example: $a = -3$ and $b = -7$ (School and Lakeside lows): $\|-3\| < \|-7\|$ but $-3 > -7$.
4	The temperature fell 1.5°F every hour from 6°F. After how many hours was it −4.5°F?	6 − 1.5h = −4.5 → 1.5h = 10.5 → 7 hours

Part 2: Elevation Profile

#	Task	Answer
5	A hiker walks from Valley to Hilltop. Find the total change in elevation and the absolute change.	420 − (−40) = +460 ft; |+460| = 460 ft
6	Find the mean elevation of all six stations. Which stations are within 50 ft of the mean?	Sum 420 − 15 + 80 − 6 + 35 − 40 = 474 → 79 ft. Within 50: School (80), Farm (35).
7	Write two different stations whose elevations have a sum of 0 if one of them moved. Explain using opposites.	Answers vary: e.g., if Farm were at +40 ft it would be the opposite of Valley (−40), so the sum would be 0.
8	Explain why $\|a - b\|$ gives the distance between two elevations no matter which is higher.	$a - b$ and $b - a$ are opposites; absolute value makes both positive, giving the same distance.

Part 3: Coordinate Coverage

#	Task	Answer
9	The radar rectangle has corners at Lakeside (−4, 2), (5, 2), (5, −4), and (−4, −4). Find its perimeter and area.	Width |5 − (−4)| = 9, height |2 − (−4)| = 6 → perimeter 30 mi, area 54 mi²
10	Which stations are outside the radar rectangle?	Hilltop (3, 4) is outside ($y = 4 > 2$). All others are inside or on the edge.
11	Design a new rectangular radar zone that covers all six stations with the least area. Give the corners and the area.	x from −4 to 5, y from −4 to 4 → corners (−4, 4), (5, 4), (5, −4), (−4, −4); area 9 × 8 = 72 mi².

Reflection

#	Prompt	Look for
R1	Summarize how the sign and the absolute value of a number each give information.	The sign gives direction (above/below, warmer/colder); the absolute value gives size or distance.
R2	How did you find side lengths on the coordinate plane without a ruler?	For points on a horizontal or vertical line, subtract the changing coordinate and take the absolute value.